

- Designing and optimizing classification and anomaly detection models with scikit-learn, XGBoost, LightGBM, and Isolation Forest.
- Developing and deploying RESTful web services using FastAPI and Flask to operationalize models as secure, scalable APIs.
- Building and maintaining scalable ETL pipelines with PySpark, Airflow, and Snowflake for large, multi-source datasets.
- Working with GenAI and LLMs, including prompt engineering and RAG frameworks using LangChain, Pinecone, and OpenAI APIs.
- Implementing MLOps practices: containerizing applications with Docker, orchestrating deployments with Kubernetes/EKS, and monitoring models with MLFlow.
- Creating interactive dashboards and visual analytics using Power BI, Tableau, and Streamlit to communicate model outputs and business KPIs.
- Leading code reviews, mentoring junior data scientists, and collaborating with cross-functional teams to align data solutions with strategic goals.
- Managing Agile deliverables, sprint planning, and ensuring continuous integration and deployment best practices are followed.

Role: Data Scientist

Responsibilities:

- Built and optimized ETL infrastructure using PySpark and Snowflake for high-volume fraud analytics workflows.
- Designed Kafka-based streaming pipelines to support real-time scoring and anomaly detection using Isolation Forest.
- Engineered feature extraction pipelines using SQL and Python for model training and batch inference.
- Implemented data validation, schema versioning, and observability with logging and automated alerts.
- Conducted unsupervised learning POCs including clustering and anomaly scoring using statistical and ML techniques.
- Delivered internal NLP tools using LangChain and FAISS to support fraud investigation with semantic search.

Accenture Solutions

October 2021 – August 2022

Role: Software Engineer

Responsibilities:

- Developed RESTful APIs using Flask and FastAPI for a regulatory dashboard, with JWT authentication and role-based access.
- Built reusable ReactJS components with Redux to support multiple user roles and workflows.
- Automated ingestion and parsing of regulatory documents with Python scripts and schema validation, reducing QA time by 40%.
- Packaged applications with Docker and configured GitLab CI/CD pipelines for automated builds and deployments.
- Participated in microservices deployments using Docker containers orchestrated on Kubernetes clusters.
- Wrote unit and integration tests with Pytest and Jest, achieving >90% test coverage.
- Participated in Agile sprints, conducted client demos, and mentored junior developers on full-stack best practices.

Agimus Solutions

October 2020 – May 2021

Role: Software Developer

Responsibilities:

- Developed backend Flask APIs for IoT device registration, sensor data ingestion, and real-time analytics.
- Secured API endpoints with OAuth2/JWT authentication; implemented JSON schema validation for payloads.
- Optimized PostgreSQL queries and indexing for faster analytics reports on large IoT datasets.
- Designed a time-series forecasting POC using moving averages and exponential smoothing for device usage trends.
- Containerized services with Docker for consistency across dev, staging, and production.
- Wrote internal API documentation and onboarding guides; maintained >90% test coverage with Pytest.

EDUCATION DETAILS:

Masters in Computer Science

May 2024

California State University San Marcos, CA

Bachelor of Engineering: Information Science & Engineering

May 2021

GSSS Institute of Engineering & Technology for Women, VTU, Mysuru

ACADEMIC PROJECTS:

RFID-Based Smart Ticketing System: Designed and deployed a prototype smart ticketing solution using RFID cards for public transportation. Implemented Python scripts to capture and validate RFID sensor data in real time, ensuring secure entry/exit logging. Integrated the system with a backend SQLite database to track passenger usage and balance updates. Developed reporting modules to generate daily, weekly, and monthly travel summaries. Tested the solution on a simulated entry gate setup to demonstrate reduced wait times and improved transaction accuracy.

ML-Based Diabetes Prediction System: Built a predictive analytics system to estimate the likelihood of diabetes onset based on patient health metrics such as BMI, glucose level, age, and family history. Utilized logistic regression and decision tree models as baselines, then incorporated deep learning techniques with Keras to compare performance. Applied polynomial feature transformation to model complex non-linear relationships and improve accuracy. Evaluated models using confusion matrix, ROC curves, and cross-validation. Visualized the final outputs with Matplotlib and presented early intervention recommendations to stakeholders for potential real-world application.